Hints: Linear algebra – 2. Vector spaces, dimension and rank

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Odd polynomials of degree at most $2m-1$ form an m-dimensional space with basis $x, x^3, \dots, x^{2m-1}$. It suffices to show that the m given polynomials lie in it and are independent.

Hint 2. Write out

$$C_{x+l-1}^{2l-1} = \frac{(x+l-1)\cdots(x+1)\,x\,(x-1)\cdots(x-l+1)}{(2l-1)!} = \frac{x(x^2-1)(x^2-4)\cdots(x^2-(l-1)^2)}{(2l-1)!}.$$

It is odd of degree exactly $2l-1$, and polynomials of distinct degrees are independent.

Problem 2

Hint 1. Expand P in the basis of binomial polynomials $C_x^k = \frac{x(x-1)\cdots(x-k+1)}{k!}$, $k = 0, \dots, n$.

Hint 2. Substituting $x = 0, 1, \dots, n$ gives a unitriangular integer system for the coefficients a_k, so by induction all $a_k \in \mathbb{Z}$. Then $n!\,C_x^k = \frac{n!}{k!}\,x(x-1)\cdots(x-k+1)$ has integer coefficients.

Problem 3

Hint 1. Let $W = \operatorname{span}\{f_1, \dots, f_n\}$, so $\dim W = n$. Differentiation $D\colon W \to \operatorname{span}\{f_1', \dots, f_n'\}$ is a linear map onto. What is its kernel?

Hint 2. $\ker D$ consists of the constant functions lying in W, so $\dim \ker D \leqslant 1$. By rank–nullity, $\dim \operatorname{span}\{f_1', \dots, f_n'\} \geqslant n-1$, and a spanning family contains a basis of its span: some $n-1$ of the f_i' are linearly independent.

Problem 4

Hint 1. Every permutation matrix has all row sums and all column sums equal to 1. Consider the space W of matrices whose row sums and column sums are all equal (to a common value). What is $\dim W$?

Hint 2. A matrix in W is determined by its upper-left $(n-1)\times(n-1)$ block and the common sum, so $\dim W = (n-1)^2 + 1$. This is an upper bound for $\dim X_n$.

Hint 3. For the lower bound, take differences $P_\sigma - P_{\sigma\circ(k\,l)}$: they give the "elementary" matrices $E_{ik} + E_{jl} - E_{il} - E_{jk}$, which span all matrices with zero row and column sums (dimension $(n-1)^2$). Add E, which is not in that span. Answer: $\dim X_n = (n-1)^2 + 1 = n^2 - 2n + 2$.

Problem 5

Hint 1. Complete u to a basis $u, e_2, \dots, e_n$. Then $A(e_j) = \lambda_j e_j + \mu_j u$ and $A(u) = cu$. The goal is to show that all λ_j are equal.

Hint 2. Apply the hypothesis to $x = e_j + e_k$ for $j \neq k \geqslant 2$ (for $\dim V = 2$ there is only one e_j and nothing to check). Comparing with $A(e_j) + A(e_k)$ and using the independence of e_j, e_k, u gives $\lambda_j = \lambda_k =: \lambda$.

Hint 3. Define φ on the basis by $\varphi(e_j) = \mu_j$ and $\varphi(u) = c - \lambda$. Both sides of $A(x) = \lambda x + \varphi(x)u$ are linear and agree on the basis.

Problem 6

Hint 1. Iterate Sylvester's inequality $\operatorname{rank}(XY) \geqslant \operatorname{rank} X + \operatorname{rank} Y - n$.

Hint 2. By induction, $\operatorname{rank}(A_1 \cdots A_k) \geqslant \sum \operatorname{rank} A_i - (k-1)n = k(n-1) - (k-1)n = n - k > 0$.

Problem 7

Hint 1. For "if": rewrite $AX + YB = AZB$ as $AX = (AZ - Y)B$ and as $YB = A(ZB - X)$. Each gives a rank inequality.

Hint 2. For "only if": equal ranks give $A = PBQ$ with P, Q invertible. You also need an invertible G with $BGB = B$: write $B = C \operatorname{diag}(E_r, 0)D$ and take $G = D^{-1}C^{-1}$.

Hint 3. Now try $X = Q^{-1}$, $Y = P$, $Z = 2Q^{-1}G$. Both sides become $2PB$.

Problem 8

Hint 1. The order is odd and the matrix is skew-symmetric, so $\det M_n = \det(-M_n^T) = -\det M_n = 0$. Thus $\operatorname{rank} M_n \leqslant 2n$.

Hint 2. For the lower bound reduce modulo 2: $M_n \equiv J - E$, where J is the all-ones matrix. A non-zero $2n$-minor modulo 2 is non-zero over $\mathbb{Q}$.

Hint 3. Over $\mathbb{F}_2$, $(J - E)x = 0$ means $x_i = \sum_j x_j$ for all i, so the kernel is spanned by $(1, \ldots, 1)$ and the rank is $2n$. (Alternatively, M_n is a circulant; compute its eigenvalues.) Answer: $\operatorname{rank} M_n = 2n$.

Problem 9

Hint 1. Express B and C as DA for suitable simple matrices D: zeroing the first row is left multiplication by $\operatorname{diag}(0, 1, \ldots, 1)$, and shifting rows up is left multiplication by the nilpotent shift J.

Hint 2. Then $(A^{-1}DA)^m = A^{-1}D^mA$, so the ranks are those of D^m. Here $D^m = D$ in (a) (rank $n - 1$ always), and $\operatorname{rank} J^m = n - m$ in (b).

Problem 10

Hint 1. Put $P = X + Y = uv^T$ and $s = v^Tu = \operatorname{tr} P$, so $P^2 = sP$. Check that $Y^2 = YP - PY$, i.e. $Y^2 = av^T - ub^T$ with $a = Yu$, $b^T = v^TY$. Note also $YPY = ab^T$.

Hint 2. Expand $Y \cdot Y^2 = Y^2 \cdot Y$ using $Y^2 = YP - PY$: $2YPY = Y^2P + PY^2 = (YP - PY)P + P(YP - PY) = s(YP - PY) = sY^2$. In vectors: $2ab^T = s(av^T - ub^T)$.

Hint 3. Show that $a = 0$ or $b = 0$. For $s = 0$ this is immediate. If $s \neq 0$ and $a, b \neq 0$, then $a(2b - sv)^T = -s\,ub^T \neq 0$ forces $Yu = a = \kappa u$. Then $v^TYu = \kappa s$ and $Y^2u = (v^Tu)a - (v^TYu)u = \kappa su - \kappa su = 0$, while $Y^2u = \kappa^2 u$; so $\kappa = 0$, a contradiction.

Hint 4. If $a = Yu = 0$, then $Y^2 = -ub^T$ and $Y^3 = -(Yu)b^T = 0$; if $b = 0$, then $Y^2 = av^T$ and $Y^3 = Y^2Y = a(v^TY) = 0$. In both cases $YPY = ab^T = 0$, hence $YXY = Y(P - Y)Y = YPY - Y^3 = 0$.

Problem 11

Hint 1.

(a) With $k = \lfloor n/2 \rfloor$, look for X with $XY = X$ of rank k and $YX = 0$. Use matrix units: $X = \sum_{i \leqslant k} e_{i,k+i}$, $Y = \sum_{i \leqslant k} e_{k+i,k+i}$.

Hint 2.

(b) Let $d = \operatorname{rank} XY - \operatorname{rank} YX$. Sylvester for YX together with $\operatorname{rank} XY \leqslant \operatorname{rank} X$ gives $d \leqslant n - \operatorname{rank} Y$. Also trivially $d \leqslant \operatorname{rank} XY \leqslant \operatorname{rank} Y$. Add the two bounds.

Problem 12

Hint 1. $(BA)^4 = B(AB)^3A = O$, so BA is nilpotent. What does this give for small n?

Hint 2. For $n \leqslant 3$, a nilpotent $n \times n$ matrix satisfies $N^3 = O$, so the answer is yes. For $n \geqslant 4$ look for a counterexample with B a shift and A diagonal.

Hint 3. In M_4, let $Se_i = e_{i-1}$ ($Se_1 = 0$) and $D = \operatorname{diag}(0, 1, 1, 1)$. With $A = D$, $B = S$, check that $(DS)^3 = O$ but $(SD)^3 e_4 = e_1$. Pad with zeros for larger n. So in general the answer is no (it holds only for $n \leqslant 3$).

Problem 13

Hint 1. $M = vw^T - wv^T$, so $Mx = (w \cdot x)v - (v \cdot x)w$ and $\operatorname{im} M \subseteq \operatorname{span}\{v, w\}$.

Hint 2. Since v, w are independent, you can solve $v \cdot x = 0$, $w \cdot x = 1$ and $v \cdot y = 1$, $w \cdot y = 0$. Then $Mx = v$ and $My = -w$, so the answer is 2.

Problem 14

Hint 1. The answers are 2 (minimum) and n (maximum). For the minimum, the natural arrangement $a_{ij} = (i-1)n + j$ has rank 2: it is the sum of two rank-one matrices.

Hint 2. Rank 1 would mean $a_{ij}a_{kl} = a_{il}a_{kj}$ for all i, j, k, l. Take a prime p with $n^2/2 < p \leqslant n^2$ (Bertrand), placed at (k, l), and pick $i \neq k$, $j \neq l$. Then p divides $a_{il}a_{kj}$, but these are entries $\leqslant n^2 < 2p$ different from p.

Hint 3. For rank n, it is enough to make the determinant odd. Reduce mod 2: there are $\lceil n^2/2 \rceil$ odd numbers, and $n \leqslant \lceil n^2/2 \rceil \leqslant n(n+1)/2$. So you can place the odd numbers on the diagonal and on some cells below it, making the matrix unit lower-triangular modulo 2.

Problem 15

Hint 1. Use idempotency to compute $(E - P - Q)P$ and $Q(E - P - Q)$. Both equal $-QP$.

Hint 2. Apply Sylvester's inequality to each product and bound $\operatorname{rank} QP$ by $\operatorname{rank} Q$, respectively by $\operatorname{rank} P$. You get $\operatorname{rank}(E - P - Q) \leqslant n - \operatorname{rank} P + \operatorname{rank} Q$ and the symmetric bound, so the answer is no.

Problem 16

Hint 1. $A - ABA = A(E - BA)$, and $\ker A \subseteq \operatorname{im}(E - BA)$: if $Ax = 0$ then $(E - BA)x = x$. Equality in Sylvester's inequality then gives $\operatorname{rank}(A - ABA) = \operatorname{rank} A + \operatorname{rank}(E - BA) - n$.

Hint 2. The same holds with A, B swapped. It remains to show $\operatorname{rank}(E - AB) = \operatorname{rank}(E - BA)$: the map $x \mapsto Bx$ sends $\ker(E - AB)$ injectively into $\ker(E - BA)$, and $y \mapsto Ay$ goes back.